

MetaGraphSci Architecture

Graph Transformer Self-Supervised Contrastive Neighbor Adaptation
for Scientific Document Classification

Project Milestone 1: Mathematical Decisions & Pipeline

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1. Problem Definition: The Low-Resource Manifold

Decision: Why target 1%, 5%, 10%, 25% labels?

Scientific domain expertise is scarce. High performance in low-label regimes is critical for specialized FoRC (Field of Research Classification).

- MetaGraphSci treats citations as semantic proxies to bridge this label gap.
- **Goal:** Optimize a shared feature graph space where structural adjacency and textual semantics are co-aligned.

2. State-of-the-Art: Representation Foundations

- **SciBERT** (Beltagy et al., 2019): Adapts BERT for scientific prose using a large-scale in-domain corpus; *it establishes the specialized vocabulary and syntactic understanding required for our foundational text encoders.*
- **SPECTER** (Cohan et al., 2020): Learns document-level embeddings by predicting citation links; *it proves that citation graphs are a powerful proxy for document semantics, driving our decision to use multi-modal relational streams.*
- **Neighborhood Contrastive Learning** (Ostendorff et al., 2022): Argues for graph-based contrastive alignment; *it provides the logic for our NCNA loss, where structural neighbors are treated as natural positive pairs to capture local consistency.*

3. State-of-the-Art: Graph Intelligence

- **Graph Transformers** (Dwivedi et al., 2022): Introduces self-attention to graph-structured data; *this enables global communication between nodes, serving as the basis for our Relation-Aware Transformer layers.*
- **NodeFormer** (Wu et al., 2022): A scalable transformer for node classification; *it utilizes a kernelized approach to handle large graphs, influencing our "Learned Selector" bottleneck for efficient neighbor pruning.*
- **SPECTER 2.0** (Ostendorff et al., 2022): Refines document representations via diverse relational types; *it justifies our decision to include explicit metadata and temporal biases within the attention mechanism.*

4. State-of-the-Art: Semi-Supervised Learning

- **FlexMatch** (Zhang et al., 2021): Introduces curriculum pseudo-labeling; *it dynamically adjusts thresholds per class, which we implement to ensure that "hard" scientific categories are not ignored during training.*
- **Calibrated SSL** (Yang et al., 2025): Addresses model over-confidence through distribution alignment; *it provides the mathematical grounding for our EMA-based thresholding and prior-matching logic.*
- **FoRC Shared Task II** (Abu Ahmad et al., 2024): The benchmark for Field of Research Classification; *it defines the standardized evaluation protocols and multi-label metrics we use to measure our model's performance.*

5. Phase 1: Meta-informed Contrastive Neighbor Adaptation

Decision: Why modify standard InfoNCE?

Standard SSL repels all batch items. In science, a neighbor is a "latent positive." We mask these to prevent "False Negative" pressure.

$$\mathcal{L}_{NCNA} = -\log \frac{\exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_{i+})/\tau)}{\exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_{i+})/\tau) + \sum_{k \in \mathcal{B} \setminus \{i\}} \omega_{ik} \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_k)/\tau)}$$

- **Topological Masking:** $\omega_{ik} = 0$ if $k \in \mathcal{N}(i)$ (cached neighbors).
- **Metadata Softening:** $\omega_{ik} = \rho$ (e.g., 0.25) if $\text{Affinity}(i, k) > 0$ (shared venue/author).

6. Deep & Cross Network (DCN) for Metadata

Decision: Handling Sparse Categorical Features

Standard MLPs struggle with sparse features like Venue or Author. DCNs capture explicit polynomial interactions in $O(d)$ time to improve classification.

$$\mathbf{x}_{l+1} = \mathbf{x}_0 \mathbf{x}_l^\top \mathbf{w}_l + \mathbf{b}_l + \mathbf{x}_l$$

- $\mathbf{x}_0, \mathbf{x}_l$: Input embedding and l -th layer output.
- $\mathbf{w}_l, \mathbf{b}_l$: Learnable weight and bias parameters.
- $\mathbf{x}_0 \mathbf{x}_l^\top$: Feature crossing term enabling $O(d)$ polynomial modeling.
- $\mathbf{x}_l$ (suffix): Residual connection ensuring gradient stability.

Combines *Cross Layers* for explicit interactions and *Deep Layers* for implicit feature learning.

7. Modality Integration: Relational Attention & Gated Fusion

Decision: Why Modality Dropout & Latent Structures?

We apply 15% modality dropout to prevent the model from relying solely on explicit citations. This forces the text encoder to internalize patterns, while **Latent Graph** discovery uncovers semantic links missing from the raw citation data.

$$\text{Attn}(i, j) = \text{Softmax} \left(\frac{(Q\mathbf{h}_i)^\top (K\mathbf{h}_j)}{\sqrt{d}} + \mathcal{B}_{topo}^{(i,j)} + \sigma(\alpha) \cdot \mathcal{A}_{latent}^{(i,j)} \right)$$

- **Structural Bias** ($\mathcal{B}_{topo}$): k -hop distance and Laplacian Eigenmaps for global topology.
- **Latent Links** ($\mathcal{A}_{latent}$): "virtual edges" between semantically related documents.
- **Gated Fusion**: Learned gate $\mathbf{g}$ balancing SciBERT features and Graph context.

The model dynamically balances explicit citations with discovered semantic relationships.

8. Phase 2: Dual-Stream Consistency & Pseudo-Labeling

Decision: Calibrating Low-Label Learning

Pseudo-labeling often biases toward "easy" majority classes. **Alignment** forces predictions to match empirical priors, while **EMA thresholds** adapt to the model's learning speed per class.

$$\mathcal{L}_{total} = \mathcal{L}_{sup} + \lambda \cdot S_u \cdot \mathcal{L}_{ce}(p_u, \hat{y}_u)$$

- $\mathcal{L}_{sup}$: Standard CE calculated on the labeled subset $\mathcal{D}_L$.
- S_u : **Selection Gate** ($S_u = 1$ if $\max p_{align} > \gamma_c$) using dynamic EMA thresholds.
- **Distribution Alignment**: Adjusts p_u by the ratio of target priors to running model expectations to prevent class bias.
- **Sharpening**: Temperature scaling produces "peakier" targets to min entropy in the unlabeled data.

Generalize in low-label regimes (1% – 10%) via calibrated pseudo-labels.

6. Integrated Training Pipeline

Require: Graph $\mathcal{G}$, Documents $\mathcal{D} = \{\mathcal{D}_L, \mathcal{D}_U\}$, Modalities $\{\mathcal{T}, \mathcal{M}, \mathcal{N}\}$, Threshold β , Weight λ

1: $\theta, \{\mathbf{w}_c\} \leftarrow$ Initialize (Transformer-QLoRA, Metadata DCN, Graph Transformer)

2: **Stage 1: Meta-informed Contrastive Neighbor Adaptation (MCNA)**

3: **for** epoch $e = 1 \dots E_{pre}$ **do**

4: $\mathcal{B} \subset \mathcal{V} \leftarrow$ Sample mini-batch; $\mathbf{z}_i \leftarrow \Phi_\theta(\mathcal{T}_i, \mathcal{M}_i, \mathcal{N}_i)$

// Generate multimodal anchor

5: $\mathbf{z}_{i+} \leftarrow \Phi_\theta(\text{Augment}(\mathcal{T}_i), \mathcal{M}_i, \mathcal{N}_i)$

// Self-supervised positive view

6: $\mu_{ik} \leftarrow \text{Affinity}(\mathcal{M}_i, \mathcal{M}_k)$

// Identify metadata-compatible candidates

7: $\mathcal{L}_{NCNA} \leftarrow -\log \frac{\exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_{i+})/\tau)}{\dots + \sum \omega_{ik} \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_k)/\tau)}$

// Mask $\mathcal{N}(i)$ to prevent false negatives

8: $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}_{NCNA}$

// Align representations with graph topology

9: **end for**

10: **Stage 2: Semi-Supervised Fine-Tuning (SSFT)**

11: **for** epoch $t = 1 \dots E_{fine}$ **do**

12: $p_L, p_U \leftarrow \Phi_\theta(\mathcal{B}_L, \mathcal{B}_U)$

// Compute posteriors for labeled and unlabeled streams

13: $\tilde{p}_U \leftarrow \text{Sharpen}(\text{Align}(p_U; \pi_{prior}))$

// Correct bias and minimize entropy

14: $T_c^{(t)} \leftarrow \beta \cdot \text{EMA}_c(t)$

// Update class-adaptive threshold via mastery

15: $M_u \leftarrow \mathbf{1}[\max \tilde{p}_u \geq T_{\hat{y}_u}^{(t)}]$

// Select high-confidence pseudo-labels $\hat{y}_u$

16: $\mathcal{L}_{total} \leftarrow \mathcal{L}_{sup}(\mathcal{B}_L) + \lambda \sum_u M_u \cdot \mathcal{L}_{ce}(\tilde{p}_u, \hat{y}_u)$

// Joint classification and consistency

17: $\theta, \{\mathbf{w}_c\} \leftarrow$ Update via $\nabla_{\theta, \mathbf{w}} \mathcal{L}_{total}$

// Optimize directional prototypes

18: **end for**

19: **return** $(\Phi_\theta, \{\mathbf{w}_c\}_{c=1}^C)$

10. Evaluation Roadmap & Standardized Metrics

Benchmarks & Scaling:

- **Standard Datasets:**
 - *OGBN-ArXiv*: 169k papers (Large-scale).
 - *FoRC*: Multidisciplinary classification.
- **Data Scarcity**: Rigorous testing at 1%, 5%, and 10% label availability.
- **Statistical Variance**: Results reported as Mean $\pm$ Std Dev across 5 seeds.

Primary Classification Metrics:

- **Accuracy & Macro F1**: Standard and imbalance-aware performance.
- **AUROC & MCC**: Probability-aware and correlation-based reliability.
- **Calibration Analysis**: Reliability diagrams to detect overconfidence.

Scientific Visualization Library

Integrated pipeline for high-dimensional manifold analysis (**UMAP/t-SNE**) and per-class error inspection using **Normalized Confusion Matrices**.